Contextualising a Hybrid ED Triage Acuity Model for the Health System in Aotearoa New Zealand

College: Whitecliffe College

AAIH8801 Applied Artificial Intelligence in Health Care - T2 2026

AAIH880_Assessment 2

Student Name: Hari Bhandari

Student ID: 20267457

Date of submission: 6 November 2026

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1. INTRODUCTION

AI systems in healthcare address two broad families of problems. Clinical issues pertain to the particular patient: identifying diabetic retinopathy from retinal images (Gulshan et al., 2016), reading screening mammograms (McKinney et al., 2020), or predicting who will need intensive care (Fernandes et al., 2020). Operational problems concern flow and scarce resources, such as predicting admission at triage (Hong et al., 2018). Emergency department (ED) triage straddles both. One acuity label determines how quickly a single patient is seen and, collectively, how the waiting room is ordered.

Assessment 1 addressed this problem. Using 418,100 stays after MIMIC-IV-ED (Johnson et al., 2023), I trained text-only, structured-only and hybrid models to predict the nurse-assigned Emergency Severity Index (ESI) level (Gilboy et al., 2012) from chief complaint text and initial vital signs. The tuned hybrid without race achieved a macro F1 of 0.5527, compared with 0.4020 for structured data alone and 0.3656 for text alone. Related studies use the same inputs to forecast admission and critical care (Hong et al., 2018; Raita et al., 2019). The expected benefit is more consistent prioritisation where human triage accuracy is modest (Mistry et al., 2018).

That benefit is conditional on validation. Topol (2019) argues that high-performance medicine earns trust only through testing in the settings where models actually run and are widely deployed. According to Wong et al. (2021), the Epic sepsis model's performance in peripheral review was significantly poorer than that of its developer. Assessment 1 showed a smaller version of the same lesson: the model with the best macro F1 missed 30.4% of true ESI 1-2 patients under its default decision rule, compared with 22.3% for the default hybrid. Better classification did not mean safer triage.

2. Relevance to the Aotearoa New Zealand Health Context

New Zealand EDs triage with the Australasian Triage Scale (ATS), which is defined differently from ESI (Australasian College for Emergency Medicine [ACEM], 2023). The Assessment 1 model is therefore a method demonstration, not a deployable tool.

The equity problem is documented. The Waitangi Tribunal (2019) determined that the Crown violated Te Tiriti o Waitangi in the strategy and distribution of primary health care, identifying institutional racism and persistent Māori inequities. Whakamaua and Ola Manuia commit the health system to reducing outcome disparities for Pacific and Māori individuals (Ministry of Health, 2020a, 2020b), and self-reported racial discrimination is linked to worse health among New Zealand adults (Harris et al., 2006). Triage is where such inequities can compound. It is the first clinical judgement a patient meets; it depends on discretion, and any variation in how that discretion is applied disappears into a single number once recorded.

Assessment 1 offers two transferable lessons. First, subgroup auditing exposed variation that overall metrics hid. At a fixed 15% miss rate, 20.0% of urgent patients aged 18-34 were missed against 8.9% of those aged 80 and over, and women were missed more often than men (17.0% against 13.2%), a gap that survived age standardisation. A similar audit by ethnicity, with Māori and Pacific patients reported separately, could show whether nurse-assigned acuity varies in ways that warrant investigation. Second, outcomes can test labels. Urgent patients the models missed were admitted less often (44.7%) than those it flagged (61.8%), a comparison local admission data could repeat by ethnicity.

The limits are as important. MIMIC-IV's race categories contain no Māori group, and Pacific peoples sit inside a combined "Asian/Pacific Islander" category of 1,291 urgent patients. That aggregation would hide the variation a New Zealand audit most needs to see. Culturally responsive care is a harder claim still. The model reads vital signs and complaint text; it cannot register whānau context, the role of support people, or how communication norms shape the way a complaint is voiced. Curtis et al. (2019) define cultural safety as an outcome rated by patients and necessitating a power transfer, which no accuracy score achieves. A triage model can flag variation.

3. Challenges and Risks

Missingness in Assessment 1 was informative, not random. A missing temperature, a missing respiratory rate and an empty pain score all pushed predictions toward ESI 1, probably because the sickest patients are triaged before a full set of observations is taken. A New Zealand ED with different documentation habits, or a redesigned electronic triage form, could weaken or reverse the signal. If completeness also varied by ethnicity, rurality or facility, the model would carry that pattern into its scores unnoticed.

Bias and unfairness is harder to remove than to measure. Labels were nurse-assigned, so a gap against the label is not necessarily a gap against clinical need (Obermeyer et al., 2019; Rajkomar et al., 2018). Dropping race narrowed the Black-White and Hispanic-White under-triage gaps without lowering macro F1, yet sex and age gaps persisted: omitting an attribute does not remove its correlates. A tension follows. Race-based adjustments in clinical algorithms have been criticised for embedding unjustified assumptions (Vyas et al., 2020), so ethnicity should not drive a New Zealand triage score. Yet ethnicity must stay in the dataset, recorded under the national ethnicity data protocols (Ministry of Health, 2017), or the model cannot be audited at all.

Clinical safety rests on the tail. In the test set, 4,434 admitted patients (5.3%) were predicted non-urgent, and ESI 5 had only 220 stays, so its estimates were unstable. Māori and Pacific subgroups in a single New Zealand ED would be smaller still, producing wide intervals that can make a real

disparity look like noise. Futoma et al. (2020) caution that performance in one hospital does not predict performance in another, and Assessment 1 had no temporal or external validation.

System-level risks sit around the model. Clinicians may defer to a score (automation bias) or, if a cutoff that holds under-triage near 15% flags roughly half of all patients, learn to ignore it. And someone must own the cutoff: moving it trades missed urgent patients against crowding, which is a clinical and ethical decision, not a tuning detail.

4. Ethical Considerations

New Zealand has no AI-specific legislation. Governance relies on guidance such as the Algorithm Charter (Stats NZ, 2020), health sector advice and voluntary frameworks, a gap Taiuru (2026) identified as a risk to minority communities when bias and data sovereignty are not formally safeguarded. The National Artificial Intelligence and Algorithm Expert Advisory Group (NAIAEAG) must approve Health NZ's Artificial Intelligence Policy before any tool processes Health NZ data, and requires tools to follow the A-SAGE principles: accessible, safe and effective, accountable, good value and unbiased (Health New Zealand | Te Whatu Ora, 2025). However, two significant policy gaps remain:

- **Scope exclusions**: The policy does not apply to clinical risk scores, meaning tools like a triage acuity model could operate without this oversight.
- **Decision-making authority:** Health NZ or the Ministry of Health retains final decision-making power over NAIAEAG's guidance (Health New Zealand | Te Whatu Ora, 2026), which means Māori expertise informs outcomes rather than determining them.

Te Tiriti & Data Governance Across the AI Lifecycle

Genuine partnership and participation require involving Māori from problem definition, rather than seeking consultation after a system is built. Taiuru (2026) demonstrated this through a case involving long-stalled mammography data, where a Māori advisory group, comprising kaumātua, health experts, and a Māori data steward, established cultural protocols and performed tikanga-based evaluations. Māori should shape problem formulation, feature selection and success metrics. To align with the Waitangi Tribunal (2019) framework for the health sector, AI development must adhere to five key principles: tino rangatiratanga, equity, active protection, options and partnership. Taiuru's (2026) Te Tiriti-centred governance framework (derived from the Wai 2252 report) applies these obligations across every stage of the AI lifecycle:

- **Tino Rangatiratanga**: Māori governance and decision-making authority from inception to deployment.
- **Active Protection:** Safeguards for the gathering and use of Māori data, such as unrestricted, former, and well-informed permission.

- **Mana Whakahaere:** Data stewardship that recognises Māori data as taonga.
- **Mana Motuhake:** Embedding tikanga into technical development and operational rollout.

Guidelines from Te Mana Raraunga (2018) and the CARE ethics (Carroll et al., 2020) highlight Māori governance over Māori data assets. Training or auditing models using Māori health records demands strict cultural oversight, alongside clear mechanisms to ensure the generated benefits flow directly back to Māori communities.

Operationalising Equity, Protection, and Transparency

Principles like active protection and equity must translate into concrete operational requirements:

- **Bounded Performance Metrics:** Performance should be reported by group, with a ceiling on under -triage for Māori and Pacific patients and an automatic pause if it is breached.
- **Preserving Human Options:** Patients and healthcare providers must retain the right to decline or override algorithmic recommendations, ensuring human-only triage workflows remain fully functional.
- **Impact Assessments:** Before live deployment, comprehensive AI, algorithmic, and privacy impact assessments must be completed with Māori input (Office of the Privacy Commissioner, 2023; Taiuru 2026).

Stage-by-Stage Responsible AI Integration

Responsible AI practices must be integrated into every phase of system development:

- **Transparency & Model Explainability:** Interpretability tools like TF-IDF term weights and SHAP feature importance offer valuable insights into how inputs drive predictions. Because SHAP describe the model rather than clinical physiology, clinician-facing interfaces must say so.
- **Documentation:** Standardised model cards (Mitchell et al., 2019; Taiuru, 2026) should clearly document intended use cases, training demographics, subgroup evaluations, and known failure conditions.
- **Accountability Structures:** Governance frameworks (WHO, 2021; Stats NZ, 2020) require a named clinical lead, Māori representation on oversight boards, and explicit escalation pathways.

Development, Validation, & Monitoring:

- **Development:** Built through authentic co-design using locally representative data.
- **Validation:** Rigorously evaluated through temporal, external, and ethnicity-disaggregated testing.

- **Deployment:** Run in silent/shadow mode to evaluate performance safely before influencing clinical decisions.
- **Monitoring:** Tracked continuously via drift and equity dashboards with pre-defined stopping rules for safety.

5. Critical Reflection on Using International Datasets

MIMIC-IV-ED holds visits to one Boston academic centre between 2011 and 2019 (Johnson et al., 2023). Its population differs from New Zealand's in demographic structure, disease burden and the pathways by which people reach an ED. Representativeness fails at the point that matters most here: there is no Māori category, and Pacific peoples are merged with Asian patients. A model trained on this data has no basis for claims about Māori, yet it would be applied to Māori. The harm comes from absence rather than from misuse of data. Taiuru (2026) notes that electronic health records often lack complete Māori data, so models perform poorly for Māori cohorts, and warns that synthetic data can hide missing representation. Foreign data standing in for local data does the same behind an impressive metric.

Cultural and clinical mismatches add to this. US race categories are not New Zealand ethnicity. The national protocols let people record several ethnicities and set rules for prioritising them in reporting (Ministry of Health, 2017), and a five-group race collapse cannot be mapped onto that. Labels differ too, since ESI is not ATS (ACEM, 2023). The text model learned a 1,000-term English vocabulary from US triage shorthand and has no handling of te reo Māori, Pacific languages or whānau-reported complaints.

Local data are the only route to equitable performance. A New Zealand version would be retrained on ATS-labelled ED records, with Māori data governance agreed before extraction, data hosted in Aotearoa unless the governing kaitiaki group permits offshore processing (Taiuru, 2026), then validated by ethnicity and monitored over time. The mammography case Taiuru (2026) described shows that access can take years, so the counter-argument deserves a hearing: international datasets allow methods to be built, tested and critiqued before local data are available, and the Assessment 1 methods (patient-level splitting, validation-based cutoffs, subgroup audit) transfer even if its fitted model does not.

The danger lies in transferring the trained model itself. Calibration established on Boston data (expected calibration error 0.020) would not be expected to hold. Label definitions differ, subgroup harms stay hidden, and a "validated" status earned overseas may lend false legitimacy to local deployment. Taiuru (2026) sets the test plainly: local validation and equity-focused metrics for any health AI used in New Zealand. The defensible position is narrow: keep the method, discard the model, and treat the international dataset as a rehearsal for work that Māori partners and local data will have to complete.

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Declaration of AI and Supporting Tools Use

Generative AI was used during this assessment. Claude was used to draft the report and suggest literature based on materials from different sources, and Grammarly was used for spelling, grammar and sentence flow correction. I reviewed the draft against my Assessment 1 results and the seminar materials, provided the references, and edited the text. I accept responsibility for this submission's final quality, referencing, and academic integrity. Assessment 1 results were produced by my own code and outputs. Supporting material: Assessment 1 report and figures; seminar 5 recording (Taiuru, 2026); Health NZ AI policy and NAIAEAG page; Journals; Privacy Commissioner AI guidance; assessment brief and rubric.